Radar range and velocity

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Separate a radar's range measurement from its line-of-sight velocity measurement.

SUCCESS CRITERIA - I can...

1. Use round-trip time for range.
2. Explain the information in a Doppler shift.
3. State what radial velocity leaves unknown.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Warm-up: bring back an earlier idea

Two EM waves in vacuum have frequencies f and **3** f. **Compare speeds** and wavelengths.

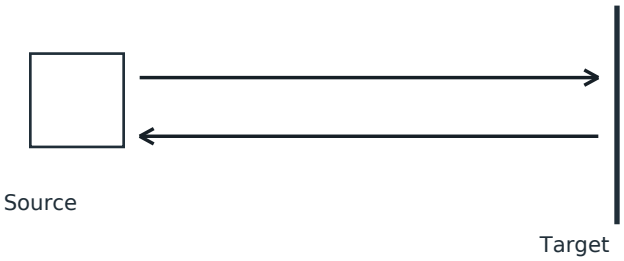
Marzano check - circle: ① ② ③

02 Worked example: explain the prediction

Reference: range = c Δt/**2**; c = **3.0×10^{8} m/s**.
Radar time is a round trip. Doppler gives line-of-sight velocity information.

Worked reasoning: A radar echo returns after **2 microseconds**. With c=**3×10^{8} m/s**, range=c Δt/**2**=**300 m**. The measured time includes the outward and return paths.

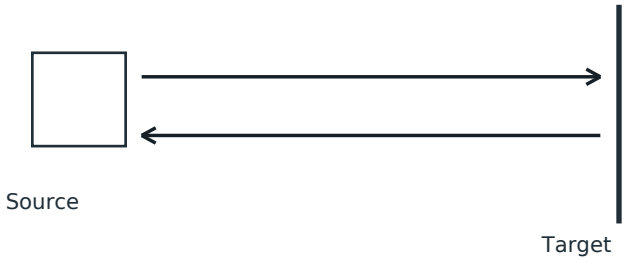
Your turn: Draw the round trip and **label the conversion 2 microseconds=2×10^{-6} s**.



Marzano check - circle: ① ② ③

03 Read and explain the diagram or equation

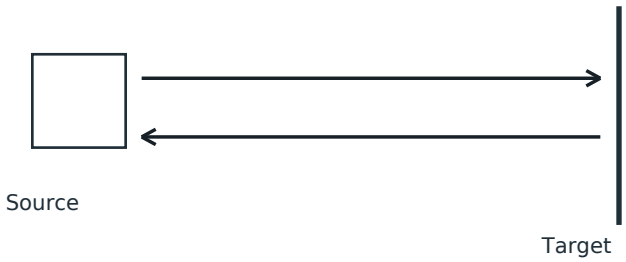
1. **Draw outgoing** and returning radar paths.
2. **Which measurement** helps range: echo delay or frequency shift?
3. Which helps radial velocity?



Marzano check - circle: ① ② ③

04 Apply the physics

A scaled radar animation uses signal speed **20 m/s** and an echo delay of **3 s**. **Label the round trip**. **Find total travel distance** and one-way range. This animation deliberately uses a slower speed than real radar.



Marzano check - circle: ① ② ③

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Continue down each column.

05 Explain your reasoning

An asteroid has zero measured radial velocity at one instant. Does that prove it is stationary? **Sketch another possible motion**.

Add a velocity perpendicular to the line of sight.

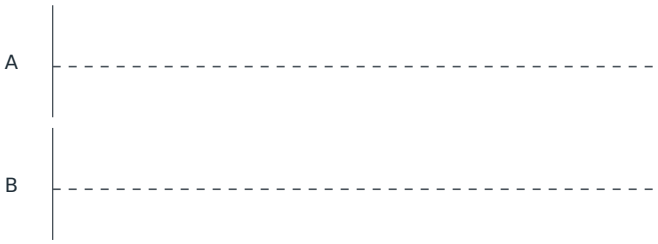


Observer Target

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

A wave has frequency **5 Hz** and wavelength **7 m**. **Find its speed**.
Label a wave sketch to **support the calculation or comparison**.

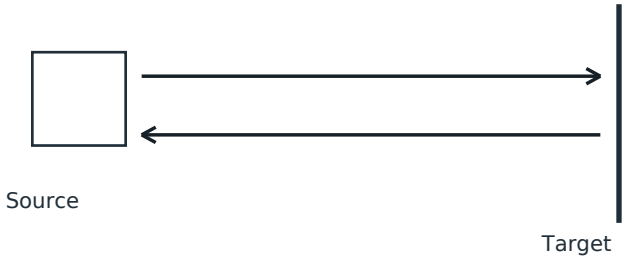


Time or position: choose and label

Marzano check - circle: 1 2 3

07 Exit check

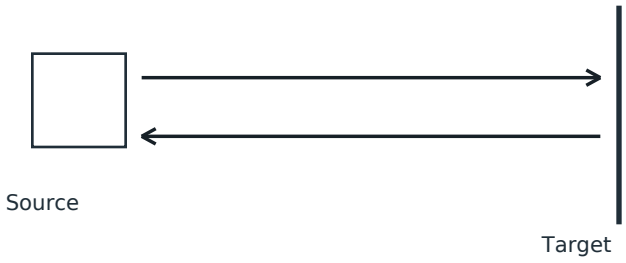
An echo’s out-and-back distance represents **1000 m**. **How far** is the target?



Marzano check - circle: 1 2 3

08 STRETCH: test the prediction

Echo times change from **4** to **3 microseconds**. **Calculate both ranges** and state whether the target is approaching or receding.



Marzano check - circle: 1 2 3